

Safety data sheet

Page: 1/14

BASF Safety data sheet according to UN GHS 4th rev.

Date / Revised: 25.10.2022

Version: 3.0

Product: **Kolliphor® CSS**

(ID no. 30554486/SDS_GEN_00/EN)

Date of print 16.05.2026

1. Identification

Product identifier

Kolliphor® CSS

Chemical name: Alcohol sulfate mono-C16-18, sodium salt

CAS Number: 68955-20-4

Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: pharmaceutical excipient

Details of the supplier of the safety data sheet

Company:

BASF SE

67056 Ludwigshafen

GERMANY

Operating Division Nutrition and Health

Telephone: +49 621 60-48434

E-mail address: EN-global-safety-data@basf.com

Emergency telephone number

International emergency number:

Telephone: +49 180 2273-112

2. Hazards Identification

Classification of the substance or mixture

According to UN GHS criteria

Flam. Sol. 2

Safety data sheet according to UN GHS 4th rev.

Date / Revised: 25.10.2022

Version: 3.0

Product: **Kolliphor® CSS**

(ID no. 30554486/SDS_GEN_00/EN)

Date of print 16.05.2026

Acute Tox. 5 (oral)
 Skin Corr./Irrit. 2
 Eye Dam./Irrit. 1
 STOT SE 3 (irritating to respiratory system)
 Aquatic Acute 2
 Aquatic Chronic 3

Specific Concentration Limits According to UN GHS Criteria

Eye Dam./Irrit. 2: 20 - 55 %
 Eye Dam./Irrit. 1: > 55 %
 Skin Irrit. 2: > 55 %

For the classifications not written out in full in this section the full text can be found in section 16.

Label elements

Globally Harmonized System (GHS)

Pictogram:



Signal Word:
 Danger

Hazard Statement:

H228	Flammable solid.
H318	Causes serious eye damage.
H315	Causes skin irritation.
H303	May be harmful if swallowed.
H335	May cause respiratory irritation.
H412	Harmful to aquatic life with long lasting effects.
H401	Toxic to aquatic life.

Precautionary Statements (Prevention):

P210	Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P280	Wear protective gloves and eye protection or face protection.
P271	Use only outdoors or in a well-ventilated area.
P273	Avoid release to the environment.
P261	Avoid breathing dust or fume.
P241	Use explosion-proof electrical, ventilating and lighting equipment.
P240	Ground and bond container and receiving equipment.
P264	Wash contaminated body parts thoroughly after handling.

Precautionary Statements (Response):

Safety data sheet according to UN GHS 4th rev.

Date / Revised: 25.10.2022

Version: 3.0

Product: **Kolliphor® CSS**

(ID no. 30554486/SDS_GEN_00/EN)

Date of print 16.05.2026

P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P310	Immediately call a POISON CENTER or physician.
P304 + P340	IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P302 + P352	IF ON SKIN: Wash with plenty of soap and water.
P370 + P378	In case of fire: Use water spray, dry powder, foam or carbon dioxide for extinction.
P362 + P364	Take off contaminated clothing and wash it before reuse.

Precautionary Statements (Storage):

P403 + P233	Store in a well-ventilated place. Keep container tightly closed.
P405	Store locked up.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.
------	---

Other hazardsAccording to UN GHS criteria

The product is under certain conditions capable of dust explosion.

3. Composition/Information on Ingredients**Substances**Chemical nature

Sulfuric acid, mono-C16-18-alkyl esters, sodium salts
 CAS Number: 68955-20-4
 EC-Number: 273-258-7

Hazardous ingredients (GHS)

According to UN GHS criteria

Sulfuric acid, mono-C16-18-alkyl esters, sodium salts

Content (W/W): >= 100 % - <= 100 %	Acute Tox. 5 (oral)
CAS Number: 68955-20-4	Skin Corr./Irrit. 2
EC-Number: 273-258-7	Eye Dam./Irrit. 1
	Aquatic Acute 2
	Aquatic Chronic 3
	H318, H315, H303, H412, H401
	<u>Specific concentration limit:</u>
	Eye Dam./Irrit. 2: 20 - 55 %
	Eye Dam./Irrit. 1: > 55 %
	Skin Irrit. 2: > 55 %

Safety data sheet according to UN GHS 4th rev.
Date / Revised: 25.10.2022
Product: **Kolliphor® CSS**

Version: 3.0

(ID no. 30554486/SDS_GEN_00/EN)

Date of print 16.05.2026

For the classifications not written out in full in this section the full text can be found in section 16.

Mixtures

Not applicable

4. First-Aid Measures

Description of first aid measures

First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Immediately remove contaminated clothing.

If inhaled:

Keep patient calm, remove to fresh air, seek medical attention. Immediately administer a corticosteroid from a controlled/metered dose inhaler.

On contact with eyes:

Immediately wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:

Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.

Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Extinguishing media

Suitable extinguishing media:
carbon dioxide, foam, dry powder, water spray

Additional information:

Avoid whirling up the material/product because of the danger of dust explosion.

Special hazards arising from the substance or mixture

carbon oxides, harmful vapours

The substances/groups of substances mentioned can be released in case of fire. Dust explosion hazard.

Advice for fire-fighters

Special protective equipment:

Wear self-contained breathing apparatus and chemical-protective clothing.

Safety data sheet according to UN GHS 4th rev.
Date / Revised: 25.10.2022
Product: **Kolliphor® CSS**

Version: 3.0

(ID no. 30554486/SDS_GEN_00/EN)

Date of print 16.05.2026

Further information:

Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.

6. Accidental Release Measures

Avoid the formation and build-up of dust - danger of dust explosion. Avoid dispersal of dust in the air (e.g. by clearing dusty surfaces with compressed air).

Personal precautions, protective equipment and emergency procedures

Use personal protective clothing. Information regarding personal protective measures, see section 8. Avoid dust formation. Ensure adequate ventilation. Do not breathe dust. Avoid contact with the skin, eyes and clothing.

Environmental precautions

Do not discharge into drains/surface waters/groundwater. Inform authorities in the event of product spillage to water courses or sewage systems.

Methods and material for containment and cleaning up

For small amounts: Contain with dust binding material and dispose of.

For large amounts: Sweep/shovel up.

Dispose of absorbed material in accordance with regulations. Cleaning operations should be carried out only while wearing breathing apparatus. Avoid raising dust.

7. Handling and Storage

Precautions for safe handling

Avoid dust formation. Provide exhaust ventilation if dust is formed. Avoid contact with the skin, eyes and clothing. Wear suitable protective clothing and eye/face protection.

Protection against fire and explosion:

The product is capable of dust explosion. Avoid dust formation. Prevent electrostatic charge - sources of ignition should be kept well clear - fire extinguishers should be kept handy. Use explosion-proof apparatus and fittings.

Conditions for safe storage, including any incompatibilities

Further information on storage conditions: Keep container tightly closed in a cool, well-ventilated place.

Specific end use(s)

For the relevant identified use(s) listed in Section 1 the advice mentioned in this section 7 is to be observed.

8. Exposure Controls/Personal Protection

Control parameters

Components with occupational exposure limits

| No substance specific occupational exposure limits known.

Exposure controls

Personal protective equipment

Respiratory protection:

Suitable respiratory protection for lower concentrations or short-term effect: Particle filter with medium efficiency for solid and liquid particles (e.g. EN 143 or 149, Type P2 or FFP2)

Hand protection:

Suitable chemical resistant safety gloves (EN ISO 374-1) also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1): E.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), butyl rubber (0.7 mm) etc. Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing. Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:

Tightly fitting safety goggles (splash goggles) (e.g. EN 166)

Body protection:

Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures

Handle in accordance with good industrial hygiene and safety practice. Wearing of closed work clothing is required additionally to the stated personal protection equipment. Do not breathe dust. No eating, drinking, smoking or tobacco use at the place of work. Hands and/or face should be washed before breaks and at the end of the shift. Store work clothing separately.

9. Physical and Chemical Properties

Information on basic physical and chemical properties

Form:	powder	
Colour:	white	
Odour:	odourless	
Odour threshold:	not applicable, odour not perceivable	
pH value:	9,2 (water, 10 %(m), 20 °C)	
Melting point:	> 150 °C (1.013 hPa)	
decomposition point:	approx. 232 °C (990 hPa) The substance / product decomposes.	(OECD Guideline 103)
Flash point:	not applicable, the product is a solid	
Evaporation rate:	The product is a non-volatile solid.	

Safety data sheet according to UN GHS 4th rev.

Date / Revised: 25.10.2022

Version: 3.0

Product: **Kolliphor® CSS**

(ID no. 30554486/SDS_GEN_00/EN)

Date of print 16.05.2026

Flammability:	Flammable solid.	(UN Test N.1 (ready combustible solids))
Lower explosion limit:	For solids not relevant for classification and labelling.	
Upper explosion limit:	For solids not relevant for classification and labelling.	
Vapour pressure:	negligible	
Relative density:	1,11 (20 °C)	(OECD Guideline 109)
Relative vapour density (air):	The product is a non-volatile solid.	
Solubility in water:	1,54 g/l (20 °C, pH 9,8)	(Directive 92/69/EEC, A.6)
Partitioning coefficient n-octanol/water (log Kow):	-0,44 (20 °C; pH value: approx. 9,8)	(OECD Guideline 107)
Self ignition:	Temperature: 20 °C not self-igniting	Test type: Spontaneous self-ignition at room-temperature.
Thermal decomposition:	>= 180 °C (DSC (DIN 51007))	
Viscosity, kinematic:	not applicable, the product is a solid	
Explosion hazard:	Product is not explosive, however a dust explosion could result from an air / dust mixture.	
Fire promoting properties:	Based on its structural properties the product is not classified as oxidizing.	

Other information

Self heating ability:	It is not a substance capable of spontaneous heating.	
SADT:	Not a substance/mixture liable to self-decomposition according to GHS.	
Bulk density:	150 - 250 kg/m3	(DGF-H-II 1b)
pKA:	Study scientifically not justified.	
Hygroscopy:	Non-hygroscopic	
Adsorption/water - soil:	log KOC: 3,5 - 3,9	(calculated)
	The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.	
Surface tension:	38,2 mN/m (24 °C; 1 g/l)	(Directive 92/69/EEC, A.5)
Grain size distribution	198 µm	(D50, Volumetric Distribution, other (measured))

10. Stability and Reactivity

Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals: No corrosive effect on metal.

Formation of Remarks:

flammable gases:

Forms no flammable gases in the presence of water.

Chemical stability

The product is stable if stored and handled as prescribed/indicated.

Possibility of hazardous reactions

Dust explosion hazard.

Conditions to avoid

Avoid dust formation. Avoid electro-static charge. See SDS section 7 - Handling and storage.

Incompatible materials

Substances to avoid:
strong oxidizing agents

Hazardous decomposition products

Hazardous decomposition products:

No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological Information

Information on toxicological effects

Acute toxicity

Assessment of acute toxicity:

The product has not been tested. The statement has been derived from substances/products of a similar structure or composition. Virtually nontoxic after a single skin contact. Of low toxicity after single ingestion.

Experimental/calculated data:

LD50 rat (oral): 4.010 mg/kg (other)

The product has not been tested. The statement has been derived from substances/products of a similar structure or composition. Analogous: Assessment derived from products with similar chemical character.

LD50 rat (dermal): > 2.000 mg/kg (OECD Guideline 402)

The product has not been tested. The statement has been derived from substances/products of a similar structure or composition. Analogous: Assessment derived from products with similar chemical character.

Irritation

Safety data sheet according to UN GHS 4th rev.

Date / Revised: 25.10.2022

Version: 3.0

Product: **Kolliphor® CSS**

(ID no. 30554486/SDS_GEN_00/EN)

Date of print 16.05.2026

Assessment of irritating effects:

Irritating to skin. Risk of serious damage to eyes. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Experimental/calculated data:

Skin corrosion/irritation rabbit: Irritant. (OECD Guideline 404)

The product has not been tested. The statement has been derived from substances/products of a similar structure or composition. Analogous: Assessment derived from products with similar chemical character.

Serious eye damage/irritation rabbit: Risk of serious damage to eyes. (OECD Guideline 405)

The product has not been tested. The statement has been derived from substances/products of a similar structure or composition. Analogous: Assessment derived from products with similar chemical character.

Respiratory/Skin sensitization

Assessment of sensitization:

Skin sensitizing effects were not observed in animal studies.

Experimental/calculated data:

Guinea pig maximization test guinea pig: Skin sensitizing effects were not observed in animal studies. (OECD Guideline 406)

Germ cell mutagenicity

Assessment of mutagenicity:

The substance was not mutagenic in bacteria. The substance was not mutagenic in mammalian cell culture. The substance was not mutagenic in a test with mammals. The product has not been fully tested. The statements have been derived in parts from products of a similar structure or composition.

Carcinogenicity

Assessment of carcinogenicity:

In long-term animal studies in which the substance was given in high doses by feed, a carcinogenic effect was not observed. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Reproductive toxicity

Assessment of reproduction toxicity:

The results of animal studies gave no indication of a fertility impairing effect. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Developmental toxicity

Assessment of teratogenicity:

The substance did not cause malformations in animal studies; however, toxicity to development was observed at high doses that were toxic to the parental animals. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Safety data sheet according to UN GHS 4th rev.

Date / Revised: 25.10.2022

Version: 3.0

Product: **Kolliphor® CSS**

(ID no. 30554486/SDS_GEN_00/EN)

Date of print 16.05.2026

Specific target organ toxicity (single exposure)

Assessment of STOT single:

Causes temporary irritation of the respiratory tract.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:

Adaptive effects were observed after repeated exposure in animal studies. Repeated oral exposure to large quantities may affect certain organs. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Aspiration hazard

| not applicable

12. Ecological Information

Toxicity

Assessment of aquatic toxicity:

Acutely toxic for aquatic organisms. Harmful to aquatic organisms based on long-term (chronic) toxicity study data. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Toxicity to fish:

LC50 (96 h) 5,2 mg/l, Brachydanio rerio (DIN EN ISO 7346-1, semistatic)

The ecological data given are those of the active ingredient.

Aquatic invertebrates:

EC50 (48 h) 15 mg/l, Daphnia magna (Directive 92/69/EEC, C.2, static)

The ecological data given are those of the active ingredient.

Analogous: Assessment derived from products with similar chemical character.

Aquatic plants:

EC50 (72 h) 34 mg/l (growth rate), Scenedesmus subspicatus (DIN 38412 Part 9, static)

The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.

An aqueous solution was tested.

EC10 (72 h) 19 mg/l (biomass), Scenedesmus subspicatus (DIN 38412 Part 9, static)

The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.

An aqueous solution was tested.

Microorganisms/Effect on activated sludge:

No observed effect concentration (18 h) 550 mg/l, Pseudomonas putida (Bringmann-Kuehn Test, static)

The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.

An aqueous solution was tested.

Chronic toxicity to fish:

Safety data sheet according to UN GHS 4th rev.

Date / Revised: 25.10.2022

Version: 3.0

Product: **Kolliphor® CSS**

(ID no. 30554486/SDS_GEN_00/EN)

Date of print 16.05.2026

No observed effect concentration (14 d) 1,65 mg/l, Brachydanio rerio (other, semistatic)
The details of the toxic effect relate to the nominal concentration. The ecological data given are those of the active ingredient.

Chronic toxicity to aquatic invertebrates:

No observed effect concentration (7 d) 0,204 mg/l, Ceriodaphnia dubia (Daphnia test chronic, Flow through.)

The statement of the toxic effect relates to the analytically determined concentration. The ecological data given are those of the active ingredient.

Analogous: Assessment derived from products with similar chemical character.

Assessment of terrestrial toxicity:

No effects at the highest test concentration.

Soil living organisms:

LC0 (14 d) > 1.000 mg/kg, Eisenia foetida (other, artificial soil)

The ecological data given are those of the active ingredient.

An aqueous solution was tested.

Terrestrial plants:

EC0 (14 d) > 1.000 mg/l, Brassica rapa (Screening test)

The ecological data given are those of the active ingredient.

An aqueous solution was tested.

Other terrestrial non-mammals:

No data available.

Persistence and degradability

Assessment biodegradation and elimination (H₂O):

Readily biodegradable (according to OECD criteria).

Elimination information:

77 % DOC reduction (30 d) (OECD 301D; EEC 92/69, C.4-E) (aerobic, activated sludge, domestic)

Readily biodegradable (according to OECD criteria).

Assessment of stability in water:

No data available.

Study scientifically not justified.

Information on Stability in Water (Hydrolysis):

Study scientifically not justified. No data available.

Bioaccumulative potential

Assessment bioaccumulation potential:

Because of the n-octanol/water distribution coefficient (log Pow) accumulation in organisms is not to be expected.

Mobility in soil

Assessment transport between environmental compartments:

Volatility: The substance will slowly evaporate into the atmosphere from the water surface.

Adsorption in soil: Adsorption to solid soil phase is possible.

Other adverse effects

Safety data sheet according to UN GHS 4th rev.

Date / Revised: 25.10.2022

Version: 3.0

Product: **Kolliphor® CSS**

(ID no. 30554486/SDS_GEN_00/EN)

Date of print 16.05.2026

The substance is not listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

13. Disposal Considerations

Waste treatment methods

Observe national and local legal requirements.

14. Transport Information

Land transport

ADR

UN number or ID number: UN1325

UN proper shipping name: FLAMMABLE SOLID, ORGANIC, N.O.S. (ALCOHOL SULFATE C16/C18, SODIUM SALT)

Transport hazard class(es): 4.1

Packing group: III

Environmental hazards: no

Special precautions for user: Tunnel code: E

RID

UN number or ID number: UN1325

UN proper shipping name: FLAMMABLE SOLID, ORGANIC, N.O.S. (ALCOHOL SULFATE C16/C18, SODIUM SALT)

Transport hazard class(es): 4.1

Packing group: III

Environmental hazards: no

Special precautions for user: None known

Inland waterway transport

ADN

UN number or ID number: UN1325

UN proper shipping name: FLAMMABLE SOLID, ORGANIC, N.O.S. (ALCOHOL SULFATE C16/C18, SODIUM SALT)

Transport hazard class(es): 4.1

Packing group: III

Environmental hazards: no

Special precautions for user: None known

Safety data sheet according to UN GHS 4th rev.

Date / Revised: 25.10.2022

Version: 3.0

Product: **Kolliphor® CSS**

(ID no. 30554486/SDS_GEN_00/EN)

Date of print 16.05.2026

Transport in inland waterway vessel

Not evaluated

Sea transport

IMDG

UN number or ID number: UN 1325

UN proper shipping name: FLAMMABLE SOLID, ORGANIC, N.O.S. (ALCOHOL SULFATE C16/C18, SODIUM SALT)

Transport hazard class(es): 4.1

Packing group: III

Environmental hazards: no

Marine pollutant: NO

Special precautions for user: EmS: F-A; S-G

Air transport

IATA/ICAO

UN number or ID number: UN 1325

UN proper shipping name: FLAMMABLE SOLID, ORGANIC, N.O.S. (ALCOHOL SULFATE C16/C18, SODIUM SALT)

Transport hazard class(es): 4.1

Packing group: III

Environmental hazards: No Mark as dangerous for the environment is needed

Special precautions for user: None known

Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

15. Regulatory Information**Safety, health and environmental regulations/legislation specific for the substance or mixture**

If other regulatory information applies that is not already provided elsewhere in this safety data sheet, then it is described in this subsection.

16. Other Information

Any other intended applications should be discussed with the manufacturer. Corresponding occupational protection measurements must be followed.

Safety data sheet according to UN GHS 4th rev.

Date / Revised: 25.10.2022

Version: 3.0

Product: **Kolliphor® CSS**

(ID no. 30554486/SDS_GEN_00/EN)

Date of print 16.05.2026

Full text of classifications, hazard symbols and hazard statements, if mentioned in section 2 or 3:

Flam. Sol.	Flammable solids
Acute Tox.	Acute toxicity
Skin Corr./Irrit.	Skin corrosion/irritation
Eye Dam./Irrit.	Serious eye damage/eye irritation
STOT SE	Specific target organ toxicity — single exposure
Aquatic Acute	Hazardous to the aquatic environment - acute
Aquatic Chronic	Hazardous to the aquatic environment - chronic
Skin Irrit.	Skin irritation
H318	Causes serious eye damage.
H315	Causes skin irritation.
H303	May be harmful if swallowed.
H412	Harmful to aquatic life with long lasting effects.
H401	Toxic to aquatic life.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.